Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cubes, cuboids, pyramids and cones. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	4
Topic	Three Dimensional Shapes
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones
Theme	—
Curriculum objectives	• Identify the properties of cube and cuboids • Identify the properties of pyramids and cones
Previous knowledge	• Students have been taught the differences between cube and cuboid.
Lesson sequence	• Identify the properties of cube and cuboids — the teacher writes the properties of cubes and cuboids on the board; the student take notes in their notebooks • Identify the properties of pyramids and cones — the teacher draws the pyramids and the cone on the board and writes their properties; the student take them down, as in Step II
Instructional materials	• Shapes of cubes, cuboid, Pyramid, cone, cylinder and spheres.
Evaluation	• How many vertices, has a cuboid? • How many edges has a cube? • Draw a cube and cuboid then label. • Draw an hexagonal base pyramid and list the properties. • How many vertices has a tetrahedron.

Lesson objectives

1.	Identify the properties of cube and cuboids
2.	Identify the properties of pyramids and cones

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cubes, cuboids, pyramids and cones

Class: JSS1 **Subject:** Mathematics **Week:** 4

Section 1: Properties of Cubes and Cuboids

Three-dimensional shapes are shapes that have length, breadth and height. Because they have all three, they are also called solid shapes. A solid shape can be regular or irregular. Regular shapes have a tidy, fixed form — a cube of sugar, a can of drink and a match box are all regular shapes. Irregular shapes are not tidy — a stone or a tree is irregular.

In this section you will learn the properties of two regular solids: the cube and the cuboid.

Before you read the properties, settle three words that describe any solid:

- A face (or surface) is a flat or curved side of a solid.
- An edge is a line where two faces meet. It may be straight or curved.
- A vertex is a point corner where edges meet. The plural of vertex is vertices.

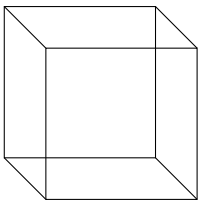
Look at fig-01. It shows a cube drawn as a see-through solid — two equal squares, one nearer to you and one farther away, with the four pairs of matching corners joined. Every face of a cube is an equal square.

A cube has these properties:

1. It is a solid shape.
2. It has 6 square faces.
3. It has 12 edges.
4. It has 8 vertices.

Count them on fig-01 if you like: six faces, twelve edges and eight vertices.

Visual



A cube drawn as a see-through solid: two equal squares, the nearer one offset down and to the right of the farther one, with the four pairs of corresponding corners joined.

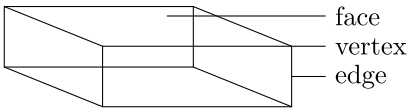
Now look at fig-02, a cuboid drawn the same way — wide and shallow. Three leader lines point to the top face, the nearer top-right vertex, and the nearer right-hand vertical edge, so you can see where the faces, vertices and edges sit.

A cuboid has these properties:

1. It has 6 faces.
2. It has 12 edges.
3. It has 8 vertices.

Notice the counts: the cube and the cuboid both have 6 faces, 12 edges and 8 vertices. What makes a cube special is that all six of its faces are equal squares; a cuboid's faces can be rectangles, so it does not have to be the same on every side.

Visual



A cuboid drawn the same way, wide and shallow, with three leader lines to the right naming the top face, the nearer top-right corner and the nearer right-hand vertical edge.

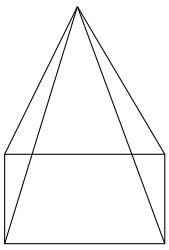
Practice

1. How many vertices does a cuboid have?
2. How many edges does a cube have?
3. Name the three words used to describe the parts of a solid, and match each one to its meaning.
4. A cube and a cuboid both have 6 faces, 12 edges and 8 vertices. What makes a cube different from a cuboid?

Section 2: Pyramids and Cones: Faces, Edges and Vertices

A pyramid gets its name from the shape of its base. Look at the triangular based pyramid in fig-03, the square based pyramid in fig-04, the pentagonal based pyramid in fig-05, and the rectangular based pyramid in fig-06. In every one, the sides rise from the base and meet at a single point at the top, called the apex. So a pyramid with a square base is called a square based pyramid, a pyramid with a triangular base is a triangular based pyramid, and so on. A triangular based pyramid has a special name of its own — a tetrahedron.

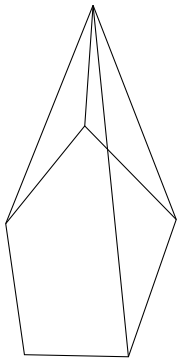
Visual



Rectangular based Pyramid

A rectangular based pyramid, drawn like the square one but on a base wider than it is tall.

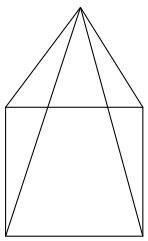
Visual



Pentagonal based Pyramid

A pentagonal based pyramid. The base is drawn as a pentagon standing on one side with its fifth vertex uppermost, and the apex above is joined to that vertex, to the two vertices either side of it, and to the lower-right vertex — four edges from the apex, the last of them crossing the face of the solid.

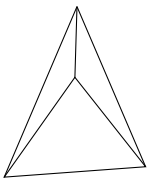
Visual



Square base Pyramid

A square base pyramid. The base is drawn as an upright rectangle and the apex above is joined to all four of its corners, so two edges run outside the rectangle and two cross it.

Visual



Triangular base Pyramid
(Tetrahedron)

A triangular base pyramid. The base triangle is drawn in perspective, its third vertex raised into the middle of the figure, and the apex above is joined to all three base vertices.

Every solid shape can be described by counting its faces (the flat surfaces), its edges (the lines where two faces meet), and its vertices (the corners). The table below counts these for the pyramids you have seen.

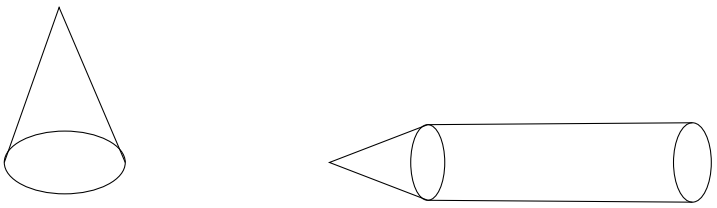
Shape	**Faces**	**Edges**	**Vertices**
Triangular based Pyramid	4	6	4
Square based Pyramid	5	8	5
Pentagonal base Pyramid	6	10	6
Rectangular based Pyramid	5	8	5

Notice the pattern: the number of faces, edges and vertices all grow together as the base gains more sides. A square based pyramid and a rectangular based pyramid each have the same counts — 5 faces, 8 edges and 5 vertices — because each base has four sides. A pentagonal based pyramid, with a five-sided base, has one more of each: 6 faces, 10 edges and 6 vertices.

A cone is different from a pyramid, as you can see in fig-07. Its base is a flat circular face, and from it a single curved surface runs up to a pointed top. A cone has:

1. Two faces — one curved face and one flat face.
2. One edge — the circular line where the flat base meets the curved surface.
3. One vertex — the pointed top.

Visual



Two drawings side by side. On the left an upright cone: an elliptical base with the two slant sides running up from the ends of the ellipse to the apex. On the right a cone lying on its side, apex at the left, its two slant sides reaching a narrow upright ellipse whose top and bottom then continue as two parallel lines to a second narrow ellipse further right.

Example. A hexagonal based pyramid has a base with six sides. How many faces, edges and vertices does it have?

Follow the pattern you saw in the table. The number of sides on the base gives the number of vertices: the base has six corners, and the apex adds one more, so the pyramid has $6 + 1 = 7$ vertices. Its faces are the six triangular side faces plus the base itself, giving $6 + 1 = 7$ faces. Each of the six base vertices joins the apex with one edge, and the base itself has six edges around it, so the pyramid has $6 + 6 = 12$ edges.

So a hexagonal based pyramid has 7 faces, 12 edges and 7 vertices.

Practice

1. A cone has how many faces, edges and vertices? Name the kind of face each one is.
2. How many vertices has a tetrahedron?
3. A square based pyramid has how many faces, edges and vertices?

Check your work

Attempt every question before you read this page.

Properties of Cubes and Cuboids

1. 8 vertices
2. 12 edges
3. Face (a flat or curved side), edge (a line where two faces meet), vertex (a point corner where edges meet).
4. All six of a cube's faces are equal squares, whereas a cuboid's faces can be rectangles.

Pyramids and Cones: Faces, Edges and Vertices

1. A cone has 2 faces (1 flat face and 1 curved face), 1 edge and 1 vertex.

   Count them straight from the cone's description: the flat circular base is one face, and the curved surface that runs up to the top is the second face. The circular line where the flat base meets the curved surface is the single edge, and the pointed top is the single vertex.
2. 4

   A tetrahedron is a triangular based pyramid. From the table, a triangular based pyramid has 4 vertices.
3. 5 faces, 8 edges, 5 vertices

   Read the row for the square based pyramid in the table: it has 5 faces, 8 edges and 5 vertices.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-868f13ef	**Date**	
Subtopic	Three dimensional figures: properties of cubes, cuboids, pyramids and cones	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____